

8 A transformation T of the plane has matrix $\mathbf{M}$, where $\mathbf{M} = \begin{pmatrix} \cos \theta & 2 \cos \theta - \sin \theta \\ \sin \theta & 2 \sin \theta + \cos \theta \end{pmatrix}$.

(a) Show that T leaves areas unchanged for all values of θ . [2]

(b) Find the value of θ , where $0 < \theta < \frac{1}{2}\pi$, for which the y -axis is an invariant line of T . [4]

The matrix $\mathbf{N}$ is $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

(c) (i) Find $\mathbf{MN}^{-1}$. [2]

(ii) Hence describe fully a sequence of two transformations of the plane that is equivalent to T . [4]